MIL-R-50263A (PA)
23 June 1976
SUPERSEDING
MIL-R-50263 (MU)
1 October 1970

MILITARY SPECIFICATION

ROCKET, 2.75 INCH, COMPLETE ROUND: ASSEMBLING, PACKING AND TESTING

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings, for the assembling, packaging, packing and testing of loaded components, of' all configurations of the 2.75 Inch Rocket.

1.1.1 The term "Complete Round" as used throughout this specification refers to the end item as it is shipped to the field, whether it is completely assembled or not, but includes a fuze, warhead and motor. This includes all configurations of the 2.75 Inch Rocket. In the specific cases prior to field use where the round is completely assembled, this specification makes reference to a COMPLETELY ASSEMBLED ROUND.

1.1.2 The specific loaded components of the 2.75 Inch Rocket covered by this specification are:

Rocket Motor, 2.75 Inch Rocket	Mark 40 & Mark 4
Warhead, 2.75 Inch Rocket	HE, M151 HE, M229 Smoke, WP, M156 Practice M230 Practice XM232 Practice XM243 Flechette WDU-4A/A XM25 5 HEDP, M247 WTU-1/B
Fuze, Rocket	M423 (Loading) M427 (Loading) M431 (Loading) Proximity M429 M435 Practice

FSC: 1340

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition Standard Quality Assurance Provisions, General Specification For.

DRAWINGS

MILITARY

90-1-44	Warhead, 2.75 Inch Rocket, Smoke, WP, M156 (with burster and fuze)
2618014	Warhead, 2.75 Inch Rocket; Practice WTU-1/B
8796521	Marking and Sealing for Ammunition Fiber Containers
8796522	Marking Diagram and Sealing for Wood Packing Boxes
8882186	Warhead, 2.75 Inch Rocket, HE, M151 w/Fuze or w/o Fuze
8883478	Cont, Ammo, Fiber, M518, For Rocket Motor: 2.75 Inch, Mark 40 and Mods, and Mark 4 and Mods
8883683	Fuze, Rocket, M423 (Loading)
8883745	Fuze, Rocket, M427 (Loading)
8886811	Container, Ammo, Fiber, M523, for Warhead, 2.75 Inch Rocket, HE, M151 Smoke WP M156
9209570	Box, Packing, Ammunition for Rocket Motor: 2.75 Inch, Mark 40 Mod 1, Mark 4, Mark 40, Mod 0 and Warheads M151, M156, XM157 and XM158

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9218698	Warhead, 2.75 Inch Rocket, HE, M229 w/Fuze or w/o Fuze
9219328	2.75 Inch (see table) Spotting Warhead XM (see table) Assembly
9220803	2.75 Inch Rocket Motor Mark 40 Mod 3 Loaded Assembly
9220804	2.75 Inch Rocket Motor Mark 4 Mod 10 Loaded Assembly
9220806	Rocket, HE, 2.75 Inch: (w/Whd., XM229)
9220807	Rocket, HE, 2.75 Inch: (w/Whd., M151)
9221050	Warhead, 2.75 Inch Rocket: Practice M230
9224841	Container, Ammo, Fiber, PA26 for Warhead, 2.75 Inch Rocket HE: - M229
9224843	Box, Packing, Ammunition for Rocket Motor: 2.75 Inch, Mark 40 and Mods, Mark 4 and Mods and Warheads M229
9227966	Warhead, 2.75 Inch Rocket, Practice XM232
9230083	Container, Ammo, Fiber, PA25 for Warhead, 2.75 Inch Rocket HE, M151, M156, XM157 & XM158 and Fuze Rocket M429
9230113	Container Ammo, Fiber, PA30 for Rocket 2.75 Inch: (Warheads M151, M156, M157, XM158, XM232 and M230)
9230114	Box, Packing Ammunition for Rocket 2.75 Inch: (Warheads M151, M156, XM157, XM158, M230, XM232 and XM247)
9230115	Container, Ammo, Fiber, PA29, for Rkt., 2.75 Inch & Whd. M229 and Practice XM243
9230116	Box, Packing Ammunition for Rocket 2.75 In., (Warhead, M229, XM245, XM255 01′ XM243)

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9230377	Container, Ammo, Fiber PA27 for Rocket 2.75 In: (Warheads M151 & M156)
9230378	Container, Ammo, Fiber, PA28 for Rocket 2.75 Inch: Warhead, HE, M229 (Fuze M429)
9232308	Container, Ammo, Fiber, PA31 for Warhead, 2.75 Inch Rocket, HE, M229 and Fuze Rocket M429
9235816	Warhead, 2.75 Inch Rocket Practice: XM243
9235840	Box, Cleated plywood, packing ammunition for Rocket 2.75 Inch (Warhead M229, XM243, XM245 or XM255)
9235841	Box, cleated plywood, packing ammunition for Rocket 2.75 Inch: (Warheads M151, M156, XM157, XM158, XM232, M230 and XM247)
9235961	Container, Ammo, Fiber, PA47 for Rocket Apers 2.75 In. (W/Warhead WDU-4A/A)
9241595	Rocket, Practice, 2.75 Inch (W/Warhead XM243)
9242070	Rocket, Apers, 2.75 Inch (W/Warhead WDU-4A/A)
9242538	Rocket, Practice, 2.75 Inch (Unassembled)
9242550	Rocket, Practice, 2.75 Inch (W/Warhead M230)
9242551	Rocket, Practice, 2.75 Inch (Unassembled)
9245122	Fuze, Rocket, M431 (Loading)
9252330	Rocket, Smoke, 2.75 Inch WP (W/Warhead M156)

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9258192	Warhead, 2.75 Inch Rocket, HE, Dual Purpose, XM247 Loaded Assembly
9269136	Container, Ammo, Fiber: PA61 for Rocket 2.75 Inch Warhead M247
9271017	Rocket, HE, Dual Purpose, 2.75 Inch (w/Warhead M247)
9276649	Rocket, Practice, 2.75 Inch (w/Warhead WTU-1/B)
9276751	Packing and Marking for Container, AMMO, Fiber: PA67 for Rocket 2.75 Inch (Warhead XM245, XM255)
9297240	Rocket, Flechette, 2.75 Inch w/Warhead XM255/w/Fuze XM439 w/Motor MK40 Mod 3
9298304	Warhead, 2.75 Inch Rocket, Flechette XM255 w/Fuze XM439

3. REQUIREMENTS

3.1 Material. -Materials shall be in accordance with applicable drawings and specifications.

3.2 Completed Assemblies. -The complete rounds and warhead-fuze combinations shall comply with all of the requirements specified in applicable drawings and specifications.

3.3 Components Assembly Torque. -The rocket components shall be assembled to withstand subsequent application of the minimum assembly torque, specified on the fuzed warhead & rocket drawings (see para. 2.1), without moving.

3.4 Ballistic Test. -The completely assembled round shall be ballistically tested.

3.4.1 Erratic Behavior. -The round shall not exhibit erratic behavior during flight. The term erratic behavior is used to describe incidents that occur during field firings which might cause the lot to be suspect from a safety or performance standpoint. These include a corkscrewing trajectory, tumbling and other deficiencies which indicate the round is not performing satisfactorily.

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3.4.2 Parts Security .-The round shall not lose parts or exhibit disassembly of components.

3.4.3 Premature Detonation. -The round shall not prematurely detonate.

3.4.4 Time to First Motion. -The maximum time interval from the instant of applying the firing current to the igniter to the time the motor thrust overcomes the 200 lb. nominal restraint and the rocket commences to move shall be as follows:

Temperature °C	Max. First Motion Time (Sec.)
-54	.075
-32	.075
+46	.028
+63	.028
+66	.023

3.4.5 Lateral Deflection. -The lateral deflection of each sample round fired shall be measured at point of warhead impact or functioning. Lateral deflection is the angle the rocket-trajectory forms with the firing azimuth along a horizontal plane. It is measured in roils with left deflection reported as minus (-) and right deflection reported as plus (+). The lot sample standard deviation shall not exceed thirty (30) mils at each temperature.

3.4.6 Rocket Motor Burnout Parameters. -Each sample round shall accelerate such that the true velocity and time (from time of ignition, To) when rocket motor burnout occurs shall be within the following limits:

	Velocity (M/Sec)		Time (Sec)	
Whd.	Lower Limit	Upper Limit	Lower Limit	Upper Limit
		-54° C		
M156	536	677	1.570	1.910
M151	536	677	1.570	1.910
XM23O	536	677	1.570	1.910
XM232	536	677	1.570	1.910
WDU-4A/A	536	677	1.570	1.910
M229	426	447	1.698	1.980
XM24 3	426	447	1.698	1.980
M247	536	677	1.570	1.910
WTU-1/B	536	677	1.570	1.910

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Whd.	Velocity (M/Sec) Lower Limit	Velocity (M/Sec) Upper Limit	Time (Sec) Lower Limit	Time (Sec) Upper Limit
		+46° C		
M156	551	696	1.438	1.710
M151	551	696	1.438	1.710
XM230	551	696	1.438	1.710
XM232	551	696	1.438	1.710
WDU-4A/A	551	696	1.438	1.710
XM25 5	551	696	1.438	1.710
M229	444	463	1.476	1.598
XM243	444	463	1.476	1.598
M247	551	696	1.438	1.710
WTU-1/B	551	696	1.438	1.710
		+66° C		
M156	549	712	1.425	1.710
M151	549	712	1.425	1.710
XM230	549	712	1.425	1,710
XM23 2	549	712	1.425	1.710
WDU-4A/A	549	712	1.425	1.710
M229	442	470	1.411	1.627
XM24 3	442	470	1.411	1.627
M247	549	712	1.425	1.710
WTU-1/B	549	712	1.425	1.710
		-32° C		
XM255	539	675	1.570	1.910
		+63° C		
XM255	583	670	1.450	1.690

3.4.7 Horizontal Range. -The minimum horizontal distance traveled by each sample round from launch to impact or function shall be 460 meters.

3.4.8 WDU-4A/A (Flechette) Warhead Functioning Parameters.- The Flechette fuze (fuze is integral part of warhead) shall function during deceleration of the rocket as evidenced by the expulsion of flechettes from the warhead. The time of functioning (from To) shall be not less than 1.385 sec. or more than 1.850 sec.

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3.4.9 M429 Proximity Fuze Parameters.-The M429 shall function between ground level and 7,62 meters above ground, initiating an acceptable function of the attached warhead. The minimum functioning time (from To) shall be 1.100 seconds.

3.4.10 H. E. Warhead Function. -H.E. loaded warheads (M151 & M229) shall detonate high order prior to ground impact when mated with an M429 fuze or upon ground impact when mated with either a M423 or M427 fuze. Also, M247 HE Warhead shall function high order upon target or ground impact.

3.4.11 White Phosphorous (WP) Warhead Functioning.-The WP warhead (M156) shall detonate prior to ground impact when mated with an M429 fuze or upon ground impact when mated with either a M423 or M427 fuze. Detonation shall disperse the WP filler such that a large white cloud is produced.

3.4.12 XM255 Warhead Functioning Parameters.-With the fuze set at one-thousand (1000) meters, the warhead shall function during the flight of the rocket as evidenced by the expulsion of flechettes and three (3) functioning tracers from the warhead. The correlated nominal time of functioning (from To) shall not be less than 2.000 seconds, nor more than 2.330 seconds (see paragraph 4.5.4). The function time is the time from first. motion of the round to first motion of the ogive from the warhead.

3.4.13 Motor Blow. -There shall be no rupture of motor tube from ignition to motor burnout.

3.5 Maximum Diameter. -Each warhead and each-motor shall be capable of passing through a ring gage of 2.792 inch diameter.

3.6 Functional Straightness. -Each complete rocket shall be capable of passing through a gage 50.100 + .025 inches long, 2.7952 + .0020 inches in width or diameter, and straight within .0020 inches total.

3.7 XM255 Warhead With Fuze XM439 Charge Circuit Resistance Test .-The resistance of the charge circuit shall be between 230 ohms minimum and 3300 ohms maximum when measured with a negative DC pulse, The pulse amplitude across the 10 megohm resistor minimum shall be -5.0 VDC maximum in amplitude and 120 microsecond maximum in duration. Pulse amplitude across the 2 ohm resistor shall be -0.020 VDC maximum in amplitude and 120 microseconds in duration.

3,8 First Article Inspection. -This specification makes provisions for First Article Inspection. Requirements for the submission of First Article samples by the contractor shall be as specified in the contract.

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3.9 Workmanship. -Assembling, packaging and packing shall be accomplished in accordance with best practice. All components shall be free of any foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions .-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections .-The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. -The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Twenty-four (24) complete assemblies

4.3.2 Inspections to be Performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality Conformance Inspection

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-R-50263A (PA)

PARAGRAPH	TITLE: Rocket, 2.75 Inch, Complete Round: Assembling, Packing and Testing		SHEET 1 OF 3		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Warhead; Practice, WTU-1/B (Dwg. 9284011)				
	Examination for Defects	24		3.2	4.4.2.1
	Warhead-Fuze Combinations; (see Para. 4.4.2.2)				
	Examination for Defects	24		3.2	4.4.2.2
	Fuze Torque	24		3.3	4.5.2
	Warhead, 2.75 Inch Rocket, Smoke, W.P. M156 (with Burster and Fuze (Dwg. 90-1-49)				
	Examination for Defects	24		3.2	4.4.2.3
	Fuze Torque	24		3.3	4.5.2

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
	Rocket, 2.75 Inch, Complete Round: Assembling Packing and Testing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Warhead, Flechette, WDU-4A/A, or XM255, Prior to assembly to motor (Dwg. 67D9700, 9298304)				
	Examination for Defects	24		3.2	4.4.2.4
	Rocket, Completely Assembled Round, Prior to Insertion into Fiber Container (see 4.4.2.5)				
	Examination for Defects	24		3.2	4.4.2.5
	Torque Test	24		3.3	4.5.2
	Warhead Electrical Circuit Test (Umbilical electronic fuzes only)	24		3.7	4.5.3.1
	Warhead Electrical Circuit Test (Umbilical electronic fuzes only)	24		3.7	4.5.3.2
	Ballistic Test	24		3.4	4.5.5

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-R-50263A (PA)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Rocket, 2.75 Inch, Complete Round: Assembling, Packing and Testing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rocket, Complete Round (see Para. 4.4.2.6)				
	Examination for Defects	24		3.2	4.4.2.6
	Torque Test	24		3.3	4.5.2
	Rocket, Complete Round (see Para. 4.4.2.7)				
	Examination for Defects	24		3.2	4.4.2.7
	Warhead Electrical Circuit Test (Umbilical electronic fuzes only)	24		3.7	4.5.3.2
	Ballistic Test	24		3.4	4.5.5

NOTES:

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4 .4.1 Inspection Lot Format ion. -Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of warheads, fuzed-warheads, and complete rounds shall contain:

a. Rocket motors from one motor production lot.

b. Warheads of one type, bearing the same interfix lot number from one metal parts manufacturer.

c. Fuzes of one type, bearing the same metal parts lot interfix number from one metal parts manufacturer.

d. The quantity of rockets in each production lot shall not exceed 20,000.

4.4.2 Examination. -See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-R-50263A (PA)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.1	Warhead; Practice, WTU-1/B		1 OF 1		9284011
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Acme Thread damaged		0.40%	3.2	Visual
102	Bourrelet damaged		0.40%	3.2	Visual
103	Diameter of bourrelet, max.		0.40%	3.5	Gage
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Warhead-Fuze Combinations; M151, M229, M230, XM232, XM253/M423, M427, M429, M435		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Fuze type improper		100%	3.2	Visual
Major					
101	Fuze Torque		0.40%	3.3	4.5.2
102	Acme thread damaged		0.40%	3.2	Visual
103	Bourrelet damaged		0.40%	3.2	Visual
104	Diameter of bourrelet, max.		0.40%	3.5	Gage
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual
NOTE:	Dwg. 8882186, 9218698, 9221050, 9227966 and 9235816)				

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CLASSIFICATION OF DEFECTS & TESTS MIL-R-50263A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Warhead, 2.75 Inch Rocket, Smoke, WP, M156 (with Burster and Fuze)				90-1-44
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Fuze type improper		100%	3.2	Visual
Major					
101	Fuze Torque		0.40%	3.3	4.5.2
102	Acme thread damaged		0.40%	3.2	Visual
103	Bourrelet damaged		0.40%	3.2	Visual
104	Diameter of bourrelet, max.		0.40%	3.5	Gage
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Compound missing from base thread		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Warhead, Flechette, WDU-4A/A, or XM255, Prior to assembly to motor				AF 67D9700, 9298304
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Acme thread damaged		0.40%	3.2	Visual
102	Nose cone not properly seated, or gap excessive		0.40%	3.2	Gage
103	Pins not flush or recessed in case, or loose or missing		0.40%	3.2	Visual
104	Bourrelet damaged		0.40%	3.2	Visual
105	Diameter of bourrelet, max.		0.40%	3.5	Gage
106	Charge circuit resistance test (XM255 only)		100%	3.7	4.5.3
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Rocket, Completely Assembled Round, Prior to Insertion into Fiber Container				See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Shorting clip improperly posi·ioned		100%	3.2	Visual
Major					
101	Functional straightness		0.04%	3.6	Gage
102	Ballistic test		(a)	3.4	4.5.5
103	Assembly torque test		0.40%	3.3	4.5.2
104	Marking missing, incorrect or illegible		0.40%	3.2	Visual
105	Warhead electrical circuit test (umbilical electronic fuzes only)	50		3.7	4.5.3.1
106	Warhead electrical circuit test (umbilical electronic fuzes only)		(b)	3.7	4.5.3.2
107	Umbilical or connector loose or not properly wrapped (umbilical electronic fuzes only)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual
NOTE:	Dwg. 9220806, 9220807, 9241595, 9242550, 9242070, 9252330, 9271017, 9276649, 9297240, As Required				

NOTES:
(a) see 4.4.3.1
(b) see 4.5.3.2

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Rocket, Complete Round (When Fuze Component is Proximity M429), Prior to Insertion into Fiber Container	1 OF 1	9220806, 9220807
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Shorting Clip improperly positioned		100%	3.2	Visual
Major					
101	Functional straightness (warhead motor only)		0.04%	3.6	Gage
102	Assembly torque (warhead motor only)		0.40%	3.3	4.5.2
103	Fuze protective bag improperly sealed		0.40%	3.2	Visual
104	Marking missing, incorrect or illegible		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Rocket, Complete Round (unassembled), Prior to Insertion into Fiber Containers. (See Note a)	1 OF 1	See Below NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Shorting Clip improperly po ' ioned		100%	3.2	Visual
Major					
101	Diameter of bourrelet, ma '. (warhead-fuze)		0.04%	3.5	Gage
102	Diameter of bourrelet, max. (motor)		0.04%	3.5	Gage
103	Ballistic Test		(b)	3.4	4.5.5
104	Warhead electrical circuit test (Umbilical electronic fuzes only)		(c)	3.7	4.5.3.2
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual
NOTE:	Dwg. 8882186, 9218698, 9219328, D90-1-44, 9220803, 9220804, 9227966, 9242538, 9242551 As Required				

NOTES: (a) These inspections cover all rockets that are shipped w/warhead-fuze combination and motor in separate fiber containers.
(b) see 4.4.3.1 (c) see 4.5.3.2

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Rocket, Complete Round, Packed in Fiber Container(s), Sealed and Taped.				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tape incomplete or badly wrinkled		0.40%	3.2	Visual
102	Cuts, scuffs, gouges or checks which penetrate outer layer		0.40%	3.2	Visual
103	Metal end loose or distorted		0.40%	3.2	Manual
Minor					
201	Length of tear tab, min.		0.65%	3.2	Gage
202	Gap between body and cover in excess of 1/8 inch		0.65%	3.2	Gage
203	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
204	Contents loose		0.65%	3.2	Manual
205	Evidence of poor workmanship		0.65%	3.9	Visual
NOTE:	Dwg.8883478, 8886811, 9224841, 9230083, 9230113, 9230115, 9230377, 9230378, 9232308, 9235961, 9269136, 9276751 As Required				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Box, Packing, Prior to Sealing	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents positioned improperly		0.40%	3.2	Visual
102	Contents packed not as marked on outside of box		0.40%	3.2	Visual
Minor					
201	Foreign material in box		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual
NOTE:	Dwg. 929570, 9224843, 9230114, 9230116, 9235840, 9235841 As Required				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Box, Packing, Sealed and Strapped, Prior to Palletizing	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents exposed		0.40%	3.2	Visual
102	Contents loose		0.40%	3.2	Visual
103	Board(s) broken		0.40%	3.2	Visual
104	Hardware missing, broken or loose		0.40%	3.2	Visual/Manual
105	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Strapping improperly assembled		0.65%	3.2	Visual
203	Hardware improperly engaged		0.65%	3.2	Visual
204	Metallic seal missing, unsealed or improperly positioned		0.65%	3.2	Visual/Manual
205	Evidence of poor workmanship		0.65%	3.9	Visual
NOTE:	Dwg. 9209570, 9224843, 9230114, 9230116, 9235840, 9235841 As Required				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Boxes, Packed, Sealed, Strapped and Palletized				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Strapping missing or loose		0.40%	3.2	Visual/Manual
102	Strapping improperly located		0.40%	3.2	Visual
103	Boxes improperly stacked or positioned		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual
NOTES:	19/48/4020/1-2-5-11PA1011 19/48/4015/12PL1000.				

NOTES:

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4 .4.3 Testing.

4.4.3.1 Ballistic Test (see 3.4) .-A sample consisting of twenty-four (24) complete rounds (as fielded) shall be randomly selected from each lot and ballistically tested at a proving ground prescribed by the contracting officer. The test shall be performed as specified in 4.5.5 using equipment in accordance with 4.4.4 and the Appendix to this specification. The sample shall be divided into three equal parts and conditioned to the temperatures prescribed in 4.5.5.1.

4.4.3.1.1 Primary Round Malfunction. -The twenty-four (24) ballistic test specimen shall be monitored for the following defects:

a. Parts Security or Premature Warhead Functioning.- (See 3.4.2, 3.4.3, 3.4.8, 3.4.9, 3.4.12).

b. Short Range. (See 3.4.7).

c. Motor Blow (See 3.4.13).

If one or more defective are found, the lot represented by the sample shall be rejected.

4.4.3.1.2 Secondary Round Malfunctions.-The twenty-four (24) ballistic test specimens shall be monitored for the following defects:

a. Excessive time of first motion (See 3.4.4).

b. Excessive lateral deflection (See 3.4.5).

c. Failure to meet motor velocity/time parameters (See 3.4.6).

d. Unacceptable warhead function (See 3.4.8, 3.4.9, 3.4.10, 3.4.11, or 3.4.12).

e. Erratic behavior (See 3.4.1).

If two or more defective are found, the lot represented by the sample shall be rejected. If there is only one defective, the lot represented by the sample shall be placed in a referred status and a retest shall be conducted. The retest sample shall consist of twenty-four (24) randomly selected complete rounds from the referred lot. The retest procedure

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shall be as specified in 4.5.5. If one or more defective are found, the lot represented by the retest sample shall be rejected. If eight (8) or more tracers fail to ignite (see 3.4.12) within an XM255 twenty-four (24) round sample, the lot represented by the sample shall be rejected with no retest permitted.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "paragraph Reference/Inspection Method" column in the Tables starting with paragraph 4.4.2.1 and including paragraph 4.4.3. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test Method and Procedures

4.5.1 Materials, Components and Processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Components Assembly Torque Test (See 3.3).-A line shall be ink marked or scribed across the junction(s) to provide a means for detection of movement between the components, The minimum torque shall then be applied in the tightening direction. Observation shall then be made for movement of the components. Any discernible movement shall be regarded as a test failure (When thread sealing compound or adhesive is used on the junction(s), the minimum torque shall be applied while the compound or adhesive is wet).

4.5.2.1 Warhead-Fuze Combination Torque Test.-This test applies to para. 4,4.2.2, 101; para. 4.4.2.3, 101.

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4.5.2.2 Completely Assembled Round Torque Test. -The Warhead-Fuze junction, if applicable, and the Warhead-Motor junction shall be tested simultaneously. This test applies to para. 4.4.2.5, 103.

4.5.2.3 Complete Round Torque Test (When fuze component is Proximity M429).-This test applies to para. 4.4.2.6, 102.

4.5.3 XM255 Warhead Charge Circuit Resistance Test

4.5.3.1 Each sample warhead/fuze assembly shall be subjected to a charge circuit resistance test. Each warhead umbilical connector shall be connected to a pulsed resistance checking device. The resistance of the charge circuit shall be ascertained. Any fuzed warhead failing the requirement shall be classed defective. One or more defects shall fail the lot. The test shall be performed using equipment in accordance with 4.4.4.

4.5.3.2 Each warhead to be ballistically tested shall be subjected to a charge circuit resistance test both before and after temperature conditioning. The resistance of the charge circuit shall be ascertained. Any fuzed warhead failing the requirement shall be classed defective. One or more defects shall fail the lot. The test shall be performed using equipment in accordance with 4.4.4.

4.5.4 XM255 Warhead Function Time.-Each warhead to be ballistically tested will have its fuze set for one-thousand (1000) meters using a charger-setter as set forth in the Appendix. A hard copy of the fuze charge signature shall be made at this time. The function time is the time from first motion of the round to first motion of the ogive from the warhead. The function time as indicated on the test equipment shall be correlated between the calibrated nominal time value of the charter-setter and the nominal time value of 2.165 seconds. The times recorded for this test are for information only.

4.5.5 Ballistic Test (See 4.4.3.1)

4.5.5.1 Each complete round selected shall be weighed (nearest one hundredth kilograms) and conditioned to assure a uniform temperature throughout its entire mass at the time of firing. The twenty-four (24) round sample shall be tested at three temperatures; eight rounds at -54 degrees centigrade,

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eight rounds at +46 degrees centigrade and eight rounds at +66 degrees centigrade except the XM255 rocket shall be tested at; eight rounds at -32 degrees centigrade, eight rounds at +46 degrees centigrade and eight rounds at +63 degrees centigrade. Minimum conditioning time shall be 24 hours. During the last three hours of conditioning, the cabinet temperature shall be maintained within three degrees of the temperatures specified. Each firing shall occur within ten minutes from the time the complete round is removed from the temperature-conditioning cabinet. A complete round left out of the conditioning cabinet in excess of ten minutes prior to firing shall be placed at ambient temperature for not less than eight hours and then reconditioned to the applicable temperature. A direct current (dc) power supply shall provide a firing pulse of 2.8 to 3.2 amperes to the ignition circuit of the rocket motor. Firing of the entire sample shall be completed even if two or more defective are obtained.

4.5.5.2 Each sample round shall be fired from a pedestal mounted launcher containing a maximum of four tubes. The launcher shall be capable of charging electronic fuzes with umbilical cables and have umbilical break-away in twelve (12) inches of rocket movement with a minimum force of two (2) pounds and a maximum force of ten (10) pounds. The launcher shall be constructed such that the launch quadrant elevation (QE) can be varied from +20 degrees to -20 degrees and azimuth varied ±30 degrees. The elevation of the longitudinal axis of the launch tubes directly above the launcher pedestal shall be considered O (zero) elevation reference. The launcher tube longitudinal axis shall be in line with the centerline of the radar.

4.5.5.3 The launcher shall be so located that point detonation (PD) fuzed rockets will impact at a slant range of approximately 900 meters when launched at a QE of 0 to -20°. A QE of approximately -7 will be used unless otherwise specified by the procuring agency. The impact area shall be such that a rocket launched from this attitude shall not have a trajectory resulting in a slant range in excess of 1,200 meters.

4.5.5.4 The launcher shall have a detent release force of 200 ±20 pounds. The launcher assembly shall be so fabricated and mounted that the azimuth (The horizontal direction expressed as the angular distance, in degrees, between TRUE NORTH and the direction of the specified object and measured clockwise from true north) or its line-of-fire and quadrant elevation (QE) will remain fixed during firing of all sample rounds. The launcher azimuth and QE shall be recorded for each sample round fired.

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4.5.5.5 The wind velocity (meters\second) and direction (azimuth) at the launcher shall be recorded for each sample round fired.

4.5.5.6 A data acquisition and reduction system shall be utilized that is capable of accurately monitoring the pre-launch, launch, flight and impact (and/or detonation); record the acquired data and be capable of rapid semi-automatic reduction of pertinent data to permit quality assessment of sample rounds by measurement of round parameters.

4.5.5.6.1 A microwave unit radar shall be utilized for data acquisition. The radar's antennae pedestal shall be permanently located to permit microwave line-of-sight tracking from launch to impact (and/or detonation). The horizontal distance from launcher to antennae pedestal shall be forty (40) meters.

4.5.5.6.2 The radar instrumentation (and data reduction) building shall be located not further than 25 meters from the antennae pedestal. The blockhouse area shall be adequately bunkered to provide instrumentation personnel protection from possible malfunctioning rounds.

4.5.5.6.3 To assure proper calibration of the microwave system, a calibration station will be required. It shall be positioned approximately 170 meters from the antennae pedestal.

4.5.5.6.4 To provide accurate near-launch flight data and lock-on signal for automatic data processor (as specified in 4.5.5.6.6) an infra-red sensor (IR) shall be positioned a horizontal distance of 7.48 meters (measured from the rear of the launcher to the center of the IR) down range on the line of fire. The pulse (or signal) generated by the sensor shall be permanently recorded as specified in 4.5.5.6.6.

4.5.5.6.5 The monitored data shall be permanently recorded on magnetic tape (1.5 mil base instrumentation quality) at 152 centimeters per-second (152 cps) utilizing a tape transport with sufficient channels. The transport shall be of instrumentation quality with servo speed control, provision for use of recorded reference signal to compensate for changes in tape speed during recording, direct record frequency response of 300 KHZ @152 cps.

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4.5.5.6.6 The data recorded during the ballistic test shall be such that subsequent data reduction utilizing the ADP will provide accurate measurement (or detection) of the following parameters:

a. Fire pulse

b. Time to first motion

c. Time to range marker

d. Velocity at range marker

e. Time to motor burnout

f. Velocity at motor burnout

g. Horizontal range at motor burnout

h. Lateral deflection at motor burnout

i. Elevation at motor burnout

j. Maximum elevation ***

k. Time to maximum elevation ***

l. Horizontal range at max. elevation ***

m. Time to impact and/or detonation

n. Horizontal range from launch to impact and/or detonation.

o. Lateral deflection at impact and/or detonation

p. Elevation at impact and/or detonation

q. Time to WDU-4A/A or XM255 fuze function*

r. Elevation at time of detonation for rounds fuzed with XM429 fuze**

*Required only when Flechette rounds are tested.
**Required only when test rounds are fuzed with XM429 fuzes.
***Required only when other than launcher elevation (ordinarily not required with negative launch elevation).

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4.5.5.7 Each flight shall be monitored by a television recording system with a vidicon attached to the radar pedestal. Vidicons can be utilized at other locations as necessary.

4.5.5.8 An analog tape containing at least the following channels of information (doppler, azimuth voltage, elevation voltage, pulses of T_O, time to first motion, time to IR, function or impact pulse, wind direction and velocity), shall be available for further processing for each flight. A digital tape of each flight shall be available for further processing. A video tape of each flight shall be available for further processing.

4.5.5.9 An initial test report shall be transmitted to be received at Picatinny Arsenal, SARPA-QA-A-R within 48 hours following completion of the ballistic test. The report shall contain the following data (minimum) and conform to the format on the following page.

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DERIVED BALLISTIC DATA SHEET

DATE: 10 APRIL 1972

ITEM DESCRIPTION	PROGRAM	COMPLETE RD. LOT NO.	MOTOR LOT NO.	WARHEAD LOT NO.	FUZE LOT NO.	LAUNCHER TYPE	LAUNCHER HT
2.75" ROCKET: MK40-M151 W/fuze M423		LOP 276-42	MCA 6-64	LOP 104-7	HAM 13-4	SPEC	-8

LOCATION	PAD NO.	TEST DATE	OPERATORS	DIST. TO RANGE MARKER	VIDEO TAPE	TAPE RECORD SPEED	TIME CODE FORMAT
Hill	1	9 APRIL 1972	SMITH, JONES, ANON	4 METERS	(X) YES ( ) NO ____MM	152	

ROUND NUMBER	TIME TO 1ST MOT	TIME TO IR	VEL AT IR	TIME TO MOTOR B/O	MOTOR B/O VEL	BURNOUT RANGE	AZIMUTH AT B/O	ELEVAT. AT B/O	TIME TO IMPACT	RANGE IMPACT	AZIMUTH IMPACT	ELEVAT. IMPACT	WHD FUNC SAT	WHD FUNC UNSAT	WIND VEL.	WIND DIRECT	COND. TEMP.	RESISTENCE	WEIGHT (KG)	RADAR TRACK	FLIGHT SAT	FLIGHT UNSAT	COMMENTS
01	19	65	25	1.530	575	520	65R	-30	3.070	1140	115R	-130	X		3.1	335	-54	2.16	14.00	YES	X		
02																							
03																							
04																							
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Item Description - Complete title of item.

Program - Fill in appropriate data.

Complete Round Lot Number - As shown on the warhead or data card (LOP-276-42).

Motor Lot Number - As shown on the rocket motor tube or data card (MCA 6-64).

Warhead Lot Number - As shown on the warhead or data card (MIS 1-15).

Fuze Lot Number - As shown on the fuze body or data card (HAM 13-4).

Launcher type - State model number or type of launcher.

Launcher Q.E. - State the quadrant elevation angle at which the rounds were fired.

Location - Location of the test.

Pad Number - The pad number from which the rockets were fired.

Test Date - Date of test (9 April 1972).

Operator - Names of operating personnel.

Video Tape - Fill in

Tape Recording Speed - Recording speed shown in CMPS.

Time Code Format - So indicate.

Time to 1st Motion - Reported in milliseconds (19).

Time to IR - Reported in milliseconds (65).

Velocity at IR - Reported in meters per second (25).

Time to Motor Burnout - Reported in seconds (1.530).

Motor Burnout Velocity - Reported in meters per second (575).

Burnout Range - Reported in meters (520).

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Azimuth at Burnout - Shown in + or - mils (+65).

Elevation at Burnout - Reported in + or - meters (-30).

Time to Impact - Shown in seconds (3.070).

Range at Impact - Shown in meters (1140).

Lateral Deflection at Impact - Shown in + or - mils (+115).

Elevation at Impact - Reported in + or - meters (-130). Zero elevation reference shall be intersection of horizontal and vertical centerlines of the rocket launcher utilized for test.

Warhead Function - Indicate satisfactory or unsatisfactory, If unsatisfactory, additional information shall be included referencing specific round malfunction and/or erratic behavior.

Wind Velocity - Shown in meters per second (3.1).

Wind Direction - Shown in degrees (TRUE).

Round Condition Temperature - Shown in degrees centigrade.

Contact Resistance - Shown in Ohms (2.16).

Weight - Shown in kilograms (14.00).

Radar Track - Yes or No.

Flight - Satisfactory or unsatisfactory. If unsatisfactory, additional information shall be included referencing specific round malfunction and/or erratic behavior.

Comments - Flight irregularities, unusual conditions, unusual circumstances, electronic fuze time, or other helpful information.

5. PREPARATION FOR DELIVERY

5.1 Packaging

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5.1.1 Level A. -The complete Rounds (see 1.1.1) shall be packaged in accordance with drawings 8883478, 8886811, 9224841, 9230083, 9230113, 9230115, 9230377, 9230378, 9232308, 9235961, 9269136, and 9276751.

5.2 Packing

5.2.1 Level A. -The packaged Complete Rounds shall be packed in accordance with drawings 9209570, 9224843, 9230114, 9230116, 9235840 and 9235841.

5.3 Marking.-Unit and exterior containers shall be marked in accordance-with instructions contained in the applicable packaging and packing drawings.

6. NOTES

6.1 Intended Use .-See paragraph 1.1 of this specification.

6.2 Ordering Data. -See MIL-A-48078

6.3 Submission of Inspection Equipment Designs for Approval. -See MIL-A-48078. Submit equipment designs as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Reviewing Activity
Army-MI

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REQUIREMENTS FOR PORTABLE RC FUZE CHARGER - SETTER APPENDIX

PURPOSE:

To charge the XM433E1 or M439 Fuze for accurate time delays to firing corresponding to Canopy Height or Airburst Range to Target settings. It is to be used to charge and set one fuze at a time through an umbilical or 6 foot long coaxial cable in laboratory function acceptance tests (centrifuge) or in field Ballistic Acceptance Tests where the fuze is charged just prior to ignition and firing of the rocket motor round from a single, four (4) or nineteen (19) tube ground launcher with adapter for RC fuzes.

REQUIREMENTS:

1. The setter must sequentially apply precise and accurate current charges and voltages levels to the two capacitors, C_t then C_f.

2. The setter must compensate for capacitors' tolerances and variations within the fuze at time of charging.

3. The setter must through as many circuits as needed - e.g. sample, track, memory, summing and comparator circuits - measure the voltages on the fuze capacitors and control the final voltage charge on C_f to make fuze function conform to the fuze equations I, II, III (see Figures 1 & 2).

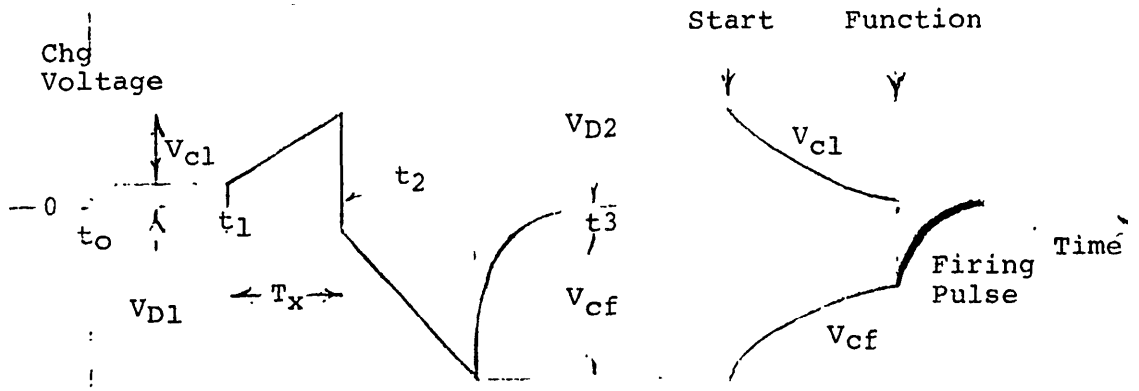


Figure 1. Fuze Charge Signature Function Signature

I Fuze Function Time Equation: $$\frac{C_f V_{cf} - C_1 V_{cl}}{C_f (V_{cl} + V_{cf})} = e^{-T_f/R_f C_s}$$

II Voltage on Cl After Time T_x: $$V_{cl} = E_1 (1 - e^{-T_x/R_x C_1})$$

III Proper Charging Requires: $$V_{cf} = V_{cl} - E_3$$

$$C_s = \frac{C_1 C_f}{C_1 + C_f}$$

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R_x = Time setting resistor which sets the range by fixing the rate of charge during 'I', thus fixes the amount of charge available to C_1.

T_x = Fixed time during which the positive charge is applied to C_1. T_x (nominal) ± 0.2%.

T_f = Time delay to function or time, R_1C_1, it takes V_{c1} to discharge from its initial set voltage to 0±75 μv.

C_1 = Timing Capacitor

C_f = Firing & Power Supply Capacitor

$t_1 - t_o$ = time delay to allow switching contacts bounce to subside.

$t_2 - t_1 = T_x$, see above and Fig. 1.

$t_3 - t_2$ = Time to charge Cf - Depends on E3 and V_{c1}.

$t_3 - t_o$ = Total time allowed to charge fuze.

Fuze Signature = Voltage vs time description at input to round during setting of fuze.

V_{c1} = Voltage level to which C_1 (timing) is charged under fixed conditions: E_1, R_x, T_x.

V_{cf} = Voltage level to which C_f is charged from a negative voltage supply E_2 until a comparator senses that $V_cf = V_{c1} - E_3$ is satisfied and stops further charging of C_f.

E_1 = Is a constant current positive voltage supply applied across $D_1 + C_1$.

E_2 = Negative D.C. power supply applied across D2 + C_f to charge C_f.

E3 = Fixed regulated reference voltage.

D1 = Steering Diode for C_1.

D2 = Steering Diode for C_f.

t_o = Time when setter receives signal to start its cycle for charging an RC Fuze.

4. The fuze charge cable from setter to fuze, assembled to round and mounted in launcher, must meet electromagnetic interference susceptibility requirements of MIL-STD-461, -462 (Cs01, Cs02, Cs06). Monitoring of fuze for timing circuitry shall be accomplished by inserting a "T" connector between the fuze umbilical and launcher adapter and making measurements directly across charge line and ground. Induced voltage shall not be greater than 0.3 volt peak. The design shall incorporate circuit elements to eliminate or reduce to tolerable levels effects from electromagnetic radiation or electrical impulse pickup.

5. The setter shall contain a fuze panel with precision range setting resistors (Rx) and switches for the selection of desired Canopy Height or Range To Target in the respective Canopy Mode (XM433E1) and Airburst Range Mode (M439 Fuze) indicated in Table 1.

6. Input power shall be 115 ± 10 volts @ 60 ± 5Hz. Internal regulated power supplies shall use the same power source. All of the following cables shall be 80 (eighty) feet long: 115V 60 Hz power cable to setter; remote control; scope sync and buffered fuze charge signature signal cables to setter; 28 VDC fire power source cable to setter. The fuze charge coaxial cable and the D.C. fire signal cable from setter to sequencer and fire box shall be 6 (six) feet long. The portable case and chassis shall be earth grounded.

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7. The charger-setter shall contain two precision RC Fuze Module Models containing all the components of the XM433E1 and M439 Fuze Modules (Picatinny Arsenal dwg's 9260685C and 9260689C) mounted on a detachable panel. All of the components for the RC Fuze Module Models shall be precision and selected for nominal values e.g. see Figure 2 where a few of the components present in the simplified diagram are specified. Additional precision capacitors with necessary switching network shall be provided for varying C_1 (timing) and C_f capacitors individually by ± 10% from their nominal values listed in Figure 2. The latter feature will provide the conditions for testing the charger's compensation feature. Switching circuitry shall also be provided: for discharging C_1 and C_f through 220 ohms loads; for simultaneously starting module timing (C_1 Discharging) and time interval counter (T.I.C.). Necessary coaxial cables for charging and monitoring of the time delay on T.I.C. shall also be provided. These models will be used in conducting checks (self test) of the calibration status of setter and operational status of ground launcher test set up.

8. The setter shall provide a self check feature for a quick, before and after a test, determination or verification of the operational and calibration status of the setter. This self-check shall incorporate a fuze set panel (R_x for the Canopy and Airburst Modes); the charger; circuits for injecting, following charging of C_t and C_f, standby one shot time delays (5 seconds for XM433E1, .1 second for M439) to the start of fuze timing; two precision RC Fuze Module Models as specified in section 7. A check on the calibration status of setter shall be performed, before and after a test, by using fuze set panel R_x inputs and the charger to charge and function XM433E1 and XM439 Precision Fuze Module Models-comparing function time to that calibration data displayed on portable charger fly leaf. In addition and in like fashion to latter check procedure, the Models' C_t and C_f capacitors shall be varied in all possible combinations of nominals and ± 10% of nominals values for the performance of a periodic check of the charger's compensation feature. Out of tolerance results shall require formal calibration.

9. The setter shall contain a procedure for determining the operational status of a ground launch set-up for performing Ballistic tests of 2.75" RKT Rounds containing RC Fuzes. The ground launch manual set-up shall consist of 115V 60 Hz. and 28 VDC power sources; switching and reset networks; the setter with set panel; camera and oscilloscope sync and buffered fuze charge signature output jacks; detachable panel mounted precision replicas of XM433E1 and M439 fuze modules (Picatinny Arsenal Dwg's 9260685C and 9260689C modified as per section 7) plus auxiliary equipment and interconnecting cables. The auxiliary equipment shall consist of a time interval counter, a memory oscilloscope for fuze charge signature/firing pulse monitoring, time recording cameras and manual sequencer for switching charge line from round to-round. The detachable fuze module umbilical cable connector shall be plugged into adapter of launcher to simulate a fuze (XM433E1 or M439). All components and cables of setter shall be enclosed in a rugged waterproof case able to withstand commercial transportation exposure.

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10. The setter, shall compliment the manual checkout/firing mode of items 8 with an automatic mode having an identical time sequence and delays of an actual helicopter pickle to firing pulse sequence. This automatic sequence should include a reset of charger and other recording instruments, a fixed delay for decay of switching transients, auto charging of fuze, automatic sync signals to recording oscilloscopes and to time-motion recording cameras and a delayed ignition pulse signal to the rocket motor. Total time from to to fire signal at rocket motor shall be 70 to 100 msec., 45 ms of this allowed for charging of fuze.

11. The charger-setter through proper adapter and cables shall be patched into a primary standard test set (PA dwg. 9260941) for periodic formal calibrations: (1) charger with R_x set panel. (2) charger + R_x set panel + two precision fuze module models where each models' C_t and C_f are set in all possible combinations of nominals and ± 10% of nominal values. Calibration results (nom- every for every station shall be posted to setter. The charger; of portable setter, settings from 50 msec to 12 seconds shall be accurate to ± 0.075%. (Fuze modules are specified in section 7).

12. The charger-setter shall, through as many filtering or grounding circuit components as it requires, preclude any extraneous charges on charge line to fuze in the time interval covering before, during or after charging of fuze. It shall provide oscilloscope sync signal and buffered fuze charge signature signal outputs and T_x output for time interval counter.

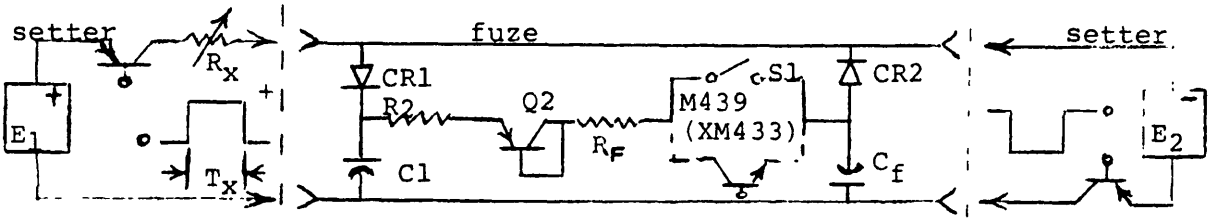


Fig. 2 SIMPLIFIED SETTER AND FUZE CKTS. DIAGRAMS

R_x = As per Table 1
CR1 (D1) = CR2 (D2)
C_1 = 6.8 MFD
C's & R's = ± 0.1%
C_f (M433) = 47 MFD
C_f (M439) = 68 MFD
R_F (M433) = 90.9K
R_F (M439) = 2.5 Meg

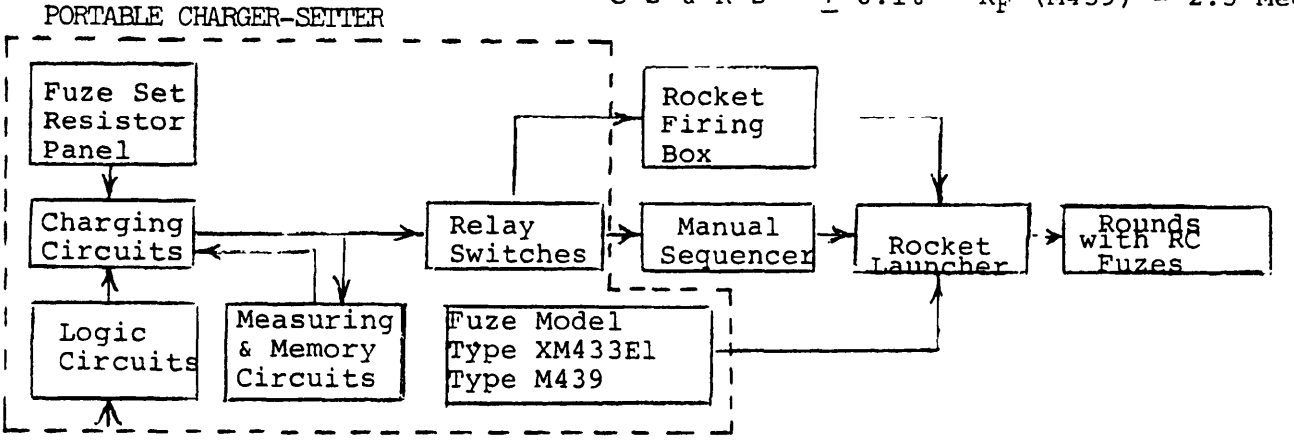


Fig. 3 GROUND LAUNCH OPERATIONAL CHECK DIAGRAM

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TABLE 1

RC FUZE, ROCKET: SETTER R_x VALUES - FUZE TIMES

CANOPY MODE				AIRBURST MODE			
Stations		R_x	M433 Fuze	Stations		R_x	M439 Fuze
Canopy Height	Added Position Resistor	Cum. Resist.	Function Time Delay **	Airburst Range To Target	Added Position Resistor	Cum. Resist.	Function Time Delay **
Meters	Ohms	Ohms	Msec (± 5% or ± 15 ms)	Meters X100	Ohms	Ohms	Seconds (±.106)
S.Q.*	open	00		5	2870	12951	1.19
	11000	20150	12.84	7.5	2320	10081	1.54
12	4530	9150	48.66	10	1520	7761	1.98
16	1500	4620	115.01	12.5	1080	6241	2.48
20	750	3120	180.08	15	780	5161	3.01
24	464	2370	244.96	17.5	590	4381	3.55
28	287	1906	311.92	20	460	3891	4.11
32	221	1619	374.74	22.5	410	3331	4.68
36	158	1398	443.56	25	320	2921	5.36
40	110	1240	508.88	27.5	270	2601	6.06
44	1130	1130	568.40	30	230	2331	6.78
Bunker				32.5	170	2101	7.52
				35	180	1931	8.21
				37.5	130	1751	9.07
				40	56	1621	9.86
				(42.5)	1565	1565	10.25

*Super Quick

Suggested Supply Source: Precision Resistor Co. Inc.
113 US Highway 22
Hillside, N.J. 07205

NOTE: All above resistor are 1/8 watts ± 1% except position resitor 1130 ± 0.1% ohms.

NOTE: All above resistors are 1/8 watts ± 0.1% ohms.

**Computer calculated nominals for above listed distance settings (stations), where $V_{cf}=V_{ct}-15.5$; E1=+16VDC; E2=-16VDC; E3=+15.5VDC; $T_x=6.000\pm0.2\%$ msec; R_x=as above.

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